

Karam Matar

Karlskrona – Sweden

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Software Engineering student at Blekinge Institute of Technology with experience in agile development and full-stack systems. I am proficient in C++, Java, and SQL, with a background in low-level software, including the development of security-focused Azure DevOps pipeline tasks. Currently serving as a Software Developer Intern and communication lead at Softhouse. I have a strong interest in algorithms, AI, and game development.

Education

Blekinge Institute of Technology

Karlskrona, Sweden

B.Sc. Software Engineering

08/2023 – 06/2026 (expected)

- Focus: systems programming, memory management, data structures, and building efficient software
- Relevant Coursework: Compiler Design, Operating Systems, Algorithms & Data Structures, Linear Algebra.

Newcastle University

Newcastle upon Tyne, United Kingdom

Exchange Semester in Computer Science (Fall 2025)

09/2025 – 01/2026 (completed)

- Focus: data analysis and cleaning datasets. basic machine learning algorithms. secure software development, and threat analysis. Systems design for games, focusing on prototyping mechanics and visual communication.
- Relevant Coursework: Computer Vision & AI, Algorithm Design and Analysis, Security Programming, Game Design.

Professional Experience

Softhouse

Karlskrona, Sweden

Software Developer Intern

Jan 2026 – Present

- Serving as the main point of contact between a team of 11 full-stack developers, the customer, and head of department to coordinate requirements and project updates.
- Working in a full-stack environment with rotating roles to develop features, handle technical documentation, and ensure clear communication within the team.
- Contributing to requirements analysis, coding, and code reviews as part of an agile development process

Trafikverket

Remote via Blekinge Institute of Technology

Software Developer Intern

Jan 2025 – May 2025

- Collaborated with a team of six to build a tool for Trafikverket that automatically checks for vulnerabilities in third-party packages
- Refactored legacy PowerShell script into JavaScript to create an Azure DevOps pipeline task that generates SBOM files in JSON format
- Took initiative in team communication by keeping absent team members informed and actively following up to help resolve blockers during development.

Blekinge Institute of Technology

Karlskrona, Sweden

Teaching Assistant (C++ Course)

Sep 2024 – Jan 2025

- Mentored first year students in C++ through lab supervision and one on one tutoring, focusing on core programming logic and best practices.
- Assisted students in debugging and mastering low-level concepts, including pointers, memory management, and object-oriented design.
- Guided students in analyzing program behavior and performance to ensure code correctness and efficiency.

Projects

Compiler from Scratch (C++): End-to-end compiler including lexer, parser, semantic analysis, CFG-based IR, TAC generation, and a custom virtual machine.

VulnerabilityChecker2.0 (PowerShell, JavaScript, Azure DevOps): Designed and implemented an Azure DevOps pipeline task for detecting vulnerabilities in third-party dependencies across multiple repositories, parsing dependency manifests, querying vulnerability data, and generating Software Bill of Materials (SBOM) files in JSON format for CI/CD workflows.

ShowcaseSERL (Next.js, React, TypeScript, Tailwind CSS): Built a full-stack web application to showcase student and researcher projects. Features include dynamic listings, detailed project views, and a kiosk mode, with a focus on

component-based architecture and clean data handling.

Death Marching Souls (C#, Unity): Developed a 3D action-platformer featuring player progression, enemy AI, boss encounters, and UI systems, implementing health management, experience-based leveling, and modular enemy behavior.

GyaGame (Java): Developed a 2D semi-open-world shooter game in Java using a provided graphics library, focusing on game logic, entity management, and player interaction.

Space Invaders (C++, SFML): Created a Space Invaders game in C++ using the SFML graphics library, implementing core game loops, collision detection, and rendering.

Skills

Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

Programming: C++, Java, Python, C#, JavaScript, TypeScript

Frameworks: React, Next.js, SvelteKit, Express.js, Node.js, Prisma

Web/Styling: HTML5, CSS, Tailwind CSS, SCSS

DevOps/Data: Azure DevOps, CI/CD, Git, MySQL, MongoDB

Game Dev: Unity, SFML

Events and Achievements

Hackathon: Hollytech Participated in a hackathon by Hollytech and won first place.